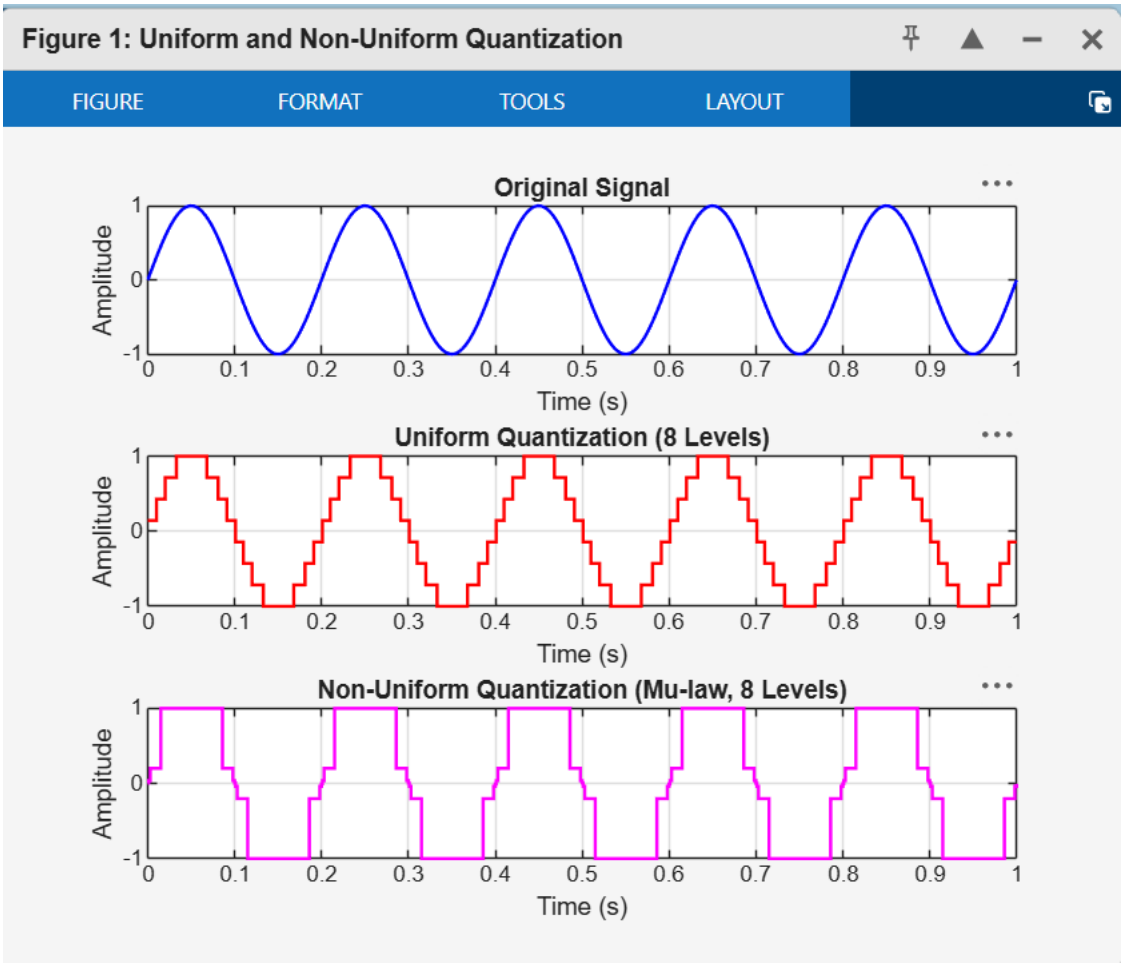
EXP 16 ADC


Figure 1: Uniform and Non-Uniform Quantization
FIGURE
FORMAT
TOOLS
LAYOUT
Original Signal
Amplitude
Time (s)
Uniform Quantization (8 Levels)
Amplitude
Time (s)
Non-Uniform Quantization (Mu-law, 8 Levels)
Amplitude
Time (s)

/MATLAB Drive/ADCexp16.m

```
clc;
clear;
close all;

% 1. INPUT SIGNAL
t = 0:0.001:1;
x = sin(2*pi*5*t);

% 2. UNIFORM QUANTIZATION (8 LEVELS)
L = 8;

% Quantize the input signal
x_uni_index = round((x + 1)*(L-1)/2);

% Reconstruct the quantized signal
x_uni = (2*x_uni_index/(L-1)) - 1;

% 3. NON-UNIFORM QUANTIZATION USING MU-LAW
mu = 255;

% Normalize and compress the signal
x_comp = sign(x).*log(1 + mu*abs(x))/log(1 + mu);
```

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```
% Reconstruct the quantized signal
x_uni = (2*x_uni_index/(L-1)) - 1;

% 3. NON-UNIFORM QUANTIZATION USING MU-LAW
mu = 255;

% Normalize and compress the signal
x_comp = sign(x).*log(1 + mu*abs(x))/log(1 + mu);

% Quantize the compressed signal
x_non_index = round((x_comp + 1)*(L-1)/2);

% Reconstruct the compressed-domain signal
x_non_comp = (2*x_non_index/(L-1)) - 1;

% Expand the signal using inverse mu-law
x_non = sign(x_non_comp).*((1 + mu).^abs(x_non_comp) - 1)/mu;

% 4. DISPLAY RESULTS
figure('Name','Uniform and Non-Uniform Quantization');

```

/MATLAB Drive/ADCexp16.m

```
37  subplot(3,1,1);
38  plot(t,x,'b','LineWidth',1.2);
39  title('Original Signal');
40  xlabel('Time (s)');
41  ylabel('Amplitude');
42  grid on;
43  axis tight;
44
45  subplot(3,1,2);
46  stairs(t,x_uni,'r','LineWidth',1.2);
47  title('Uniform Quantization (8 Levels)');
48  xlabel('Time (s)');
49  ylabel('Amplitude');
50  grid on;
51  axis tight;
52
53  subplot(3,1,3);
54  stairs(t,x_non,'m','LineWidth',1.2);
55  title('Non-Uniform Quantization (Mu-law, 8 Levels)');
56  xlabel('Time (s)');
57  ylabel('Amplitude');
58  grid on;
59  axis tight;
60
```